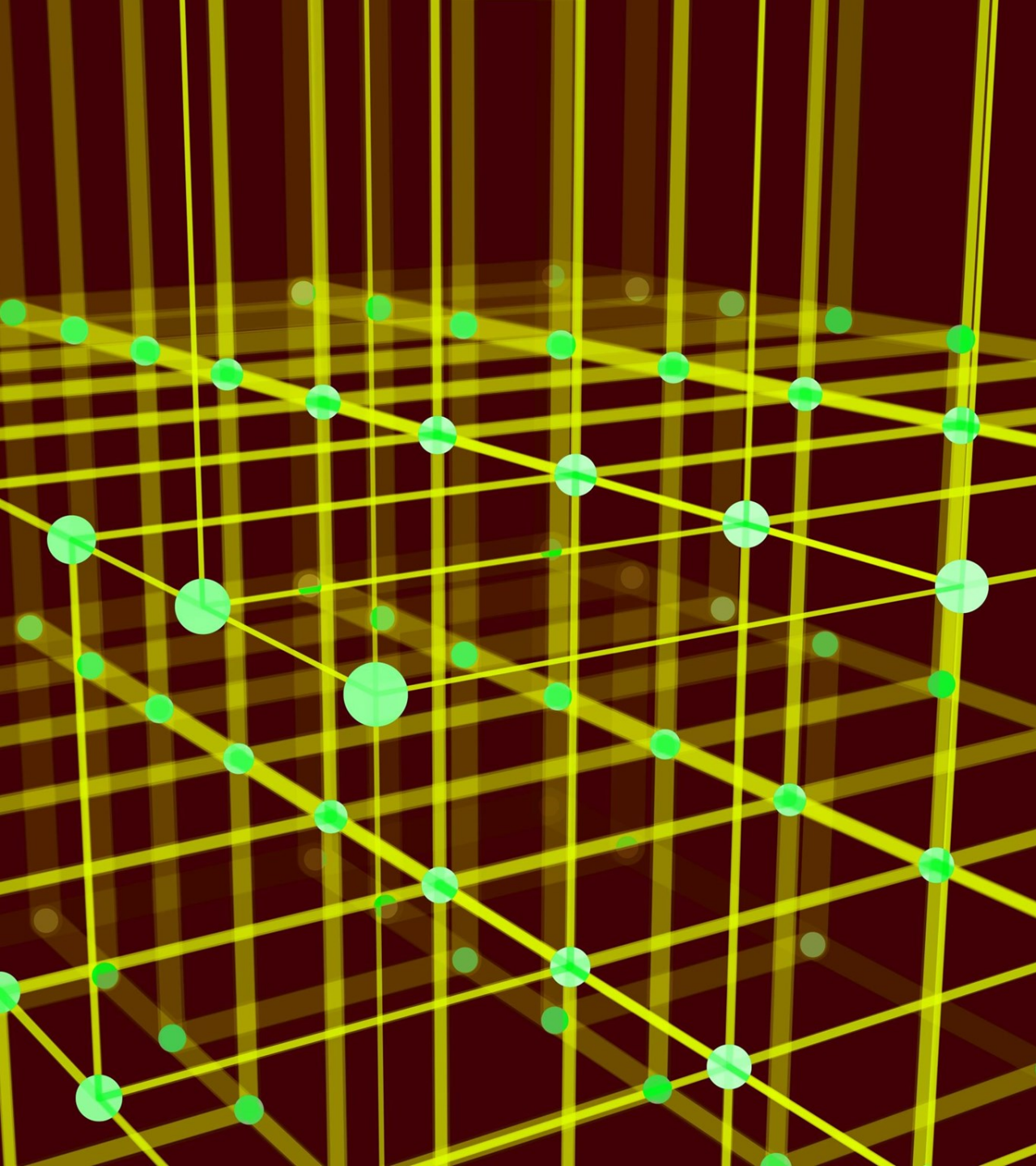




Discrete event simulation

Dr. Shankar K. Ghosh

Assistant Professor

Department of Computer Science
and Engineering

Shiv Nadar Institution of Eminence,
Delhi, NCR



A simple example

Scenario-1: An object is moving from point A to point B step-by-step. Distance between A and B is d . The object moves with a constant velocity v . In each step, the object moves by w distance. How long it will take to go from point A to point B?

Very simple right?

Making it little complex

Scenario-1.1: An object is moving from point A to point B step-by-step. Distance between A and B is d . The object moves with a constant velocity v . In alternating steps, the object moves forward by w distance and moves backward by x distance. How long it will take to go from point A to point B?

Still solvable mathematically, right? All components are deterministic!

Making it little complex

Scenario-1.2: An object is moving from point A to point B step-by-step. Distance between A and B is d . The object moves with a constant velocity v . In alternating steps, the object moves forward by a random distance w (between 1 and 10) and moves backward by another random distance x (between 1 and 5). How long it will take to go from point A to point B?

Is it easy to solve mathematically? All components are not deterministic!!

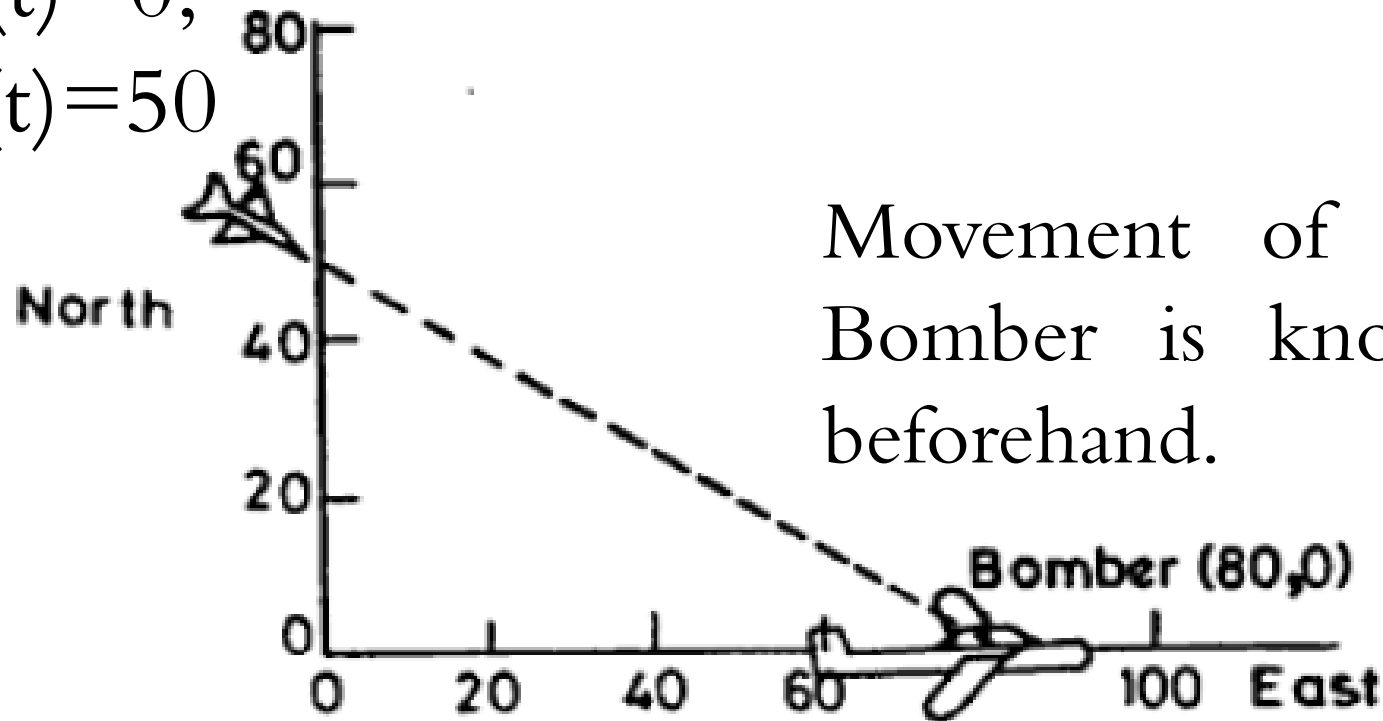
How to solve?

-Simulation

The pursuit problem

$$XF(t)=0;$$

$$YF(t)=50$$



Movement of the
Bomber is known
beforehand.

Modelling the behavior

Co-ordinates of the bomber $XB(t), YB(t)$

Time, t	0	1	2	3	4	5	6	7	8	9	10	11	12
$XB(t)$	80	90	99	108	116	125	133	141	151	160	169	179	180
$YB(t)$	0	-2	-5	-9	-15	-18	-23	-29	-28	-25	-21	-20	-17

Model assumptions

1. The target and the pursuer are flying in the same horizontal plane when the fighter first sights the bomber, and both stay in that plane. This makes the pursuit model two-dimensional.
2. The fighter's speed V_F is constant (20 kms/minute).
3. The target's path (i.e., its position as a function of time) is specified.
4. After a fixed time span Δt (every minute, in this case) the fighter changes its direction in order to point itself toward the bomber.

Updating the position of the pursuer

The distance $DIST(t)$ at a given time t between the target and the pursuer is given by

$$DIST(t) = \sqrt{(YB(t) - YF(t))^2 + (XB(t) - XF(t))^2} \quad \dots(1-1)$$

The angle θ of the line from the fighter to the target at a given time t is given by

$$\sin \theta = \frac{YB(t) - YF(t)}{DIST(t)} \quad \dots(1-2)$$

$$\cos \theta = \frac{XB(t) - XF(t)}{DIST(t)} \quad \dots(1-3)$$

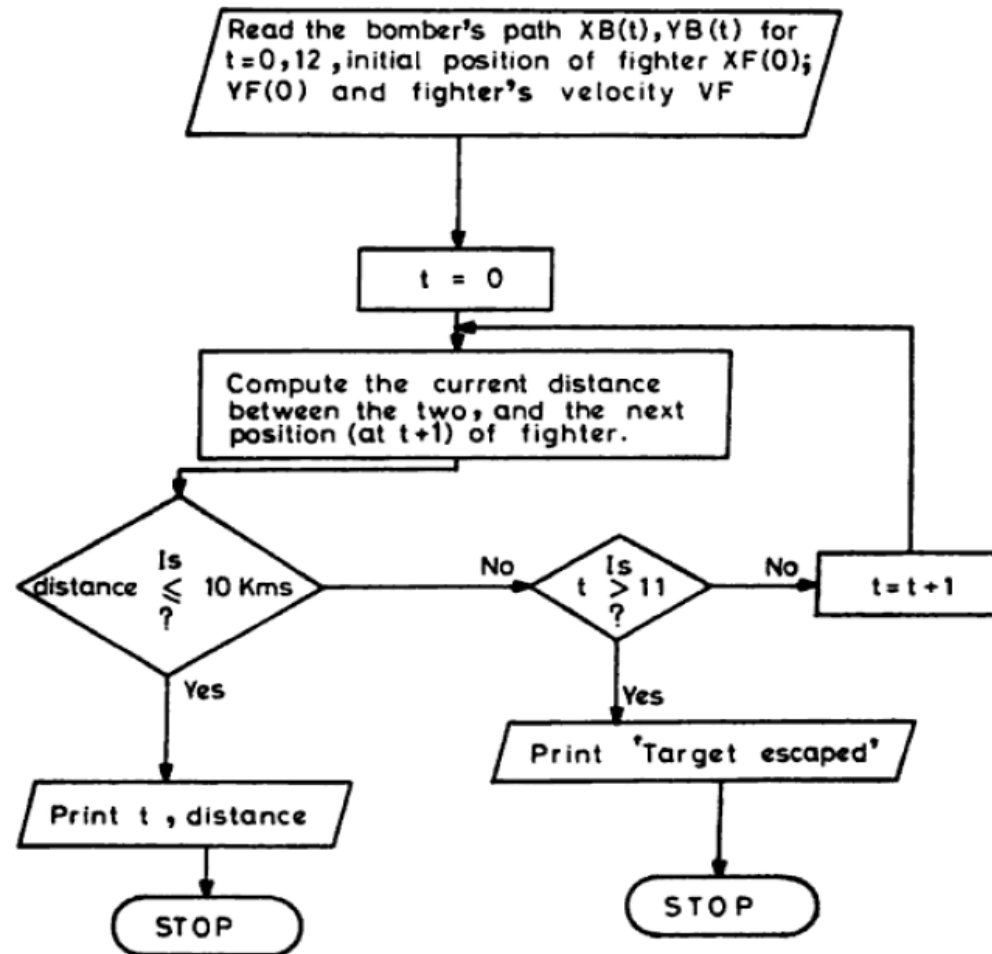
Using this value of the position of the fighter at time $(t + 1)$ is determined by

$$XF(t + 1) = XF(t) + VF \cos \theta \quad \dots(1-4)$$

$$YF(t + 1) = YF(t) + VF \sin \theta \quad \dots(1-5)$$

With these new coordinates of the pursuer, its distance from the target is again computed using Eq. (1-1). If this distance is 10 kms. or less the pursuit is over, otherwise θ is recomputed, and the process continues.

Flowchart describing the simulation



Inventory Problem

When my stock goes down to P items (called reorder point), I will order Q more items (called reorder quantity) from the wholesaler.

If the demand on any day exceeds the amount of inventory in hand, the excess represents lost sale and loss of goodwill. On the other hand, overstocking implies increased carrying cost (i.e., cost of storage, insurance, interest, deterioration etc.). Ordering too frequently will result in excessive reorder cost.

Inventory Problem: Conditions

1. There is a 3-day lag between the order and arrival. The merchandise is ordered at the end of the day and is received at the beginning of the fourth day. That is, merchandise ordered on the evening of the i^{th} day is received on the morning of the $(i+3)^{\text{rd}}$ day.
2. For each unit of inventory, the carrying cost for each night is 75 USD.
3. Each unit out of stock when ordered results into a loss of goodwill worth 2 USD per unit + loss of USD 16 net income, that would have resulted in its sale, or a total loss of USD 18 per unit. Lost sales are lost forever; they cannot be backordered.
4. Placement of each order costs USD 75 regardless of the number of units ordered.
5. The demand in a day can be for any number of units between 0 and 99, each equiprobable.
6. There is never more than one replenishment order outstanding.
7. Initially, we have 115 units on hand and no reorder outstanding.

Compare the following replenishment policies

	P (reorder point)	Q (reorder quantity)
Policy I	125	150
Policy II	125	250
Policy III	150	250
Policy IV	175	250
Policy V	175	300

Cost of Inventory policy (P, Q)

